Efficiency

Questions of efficiency

The efficiency of a machine can be worked out using the equation:

$$\textbf{Efficiency} = \frac{\text{useful energy output}}{\text{total energy input}} \times 100$$

Use this equation to answer the following questions:

1 An engine has an energy input of 100 J and gives out 27 J of useful kinetic energy. What is the efficiency of the engine?

2 A hair dryer has an energy input of 1500 J and gives out 1300 J of useful energy. What is the efficiency of the hair dryer?

3 Compare these two machines to decide which is more efficient.

i Machine A uses 1000 J of energy to do 300 J of useful work.

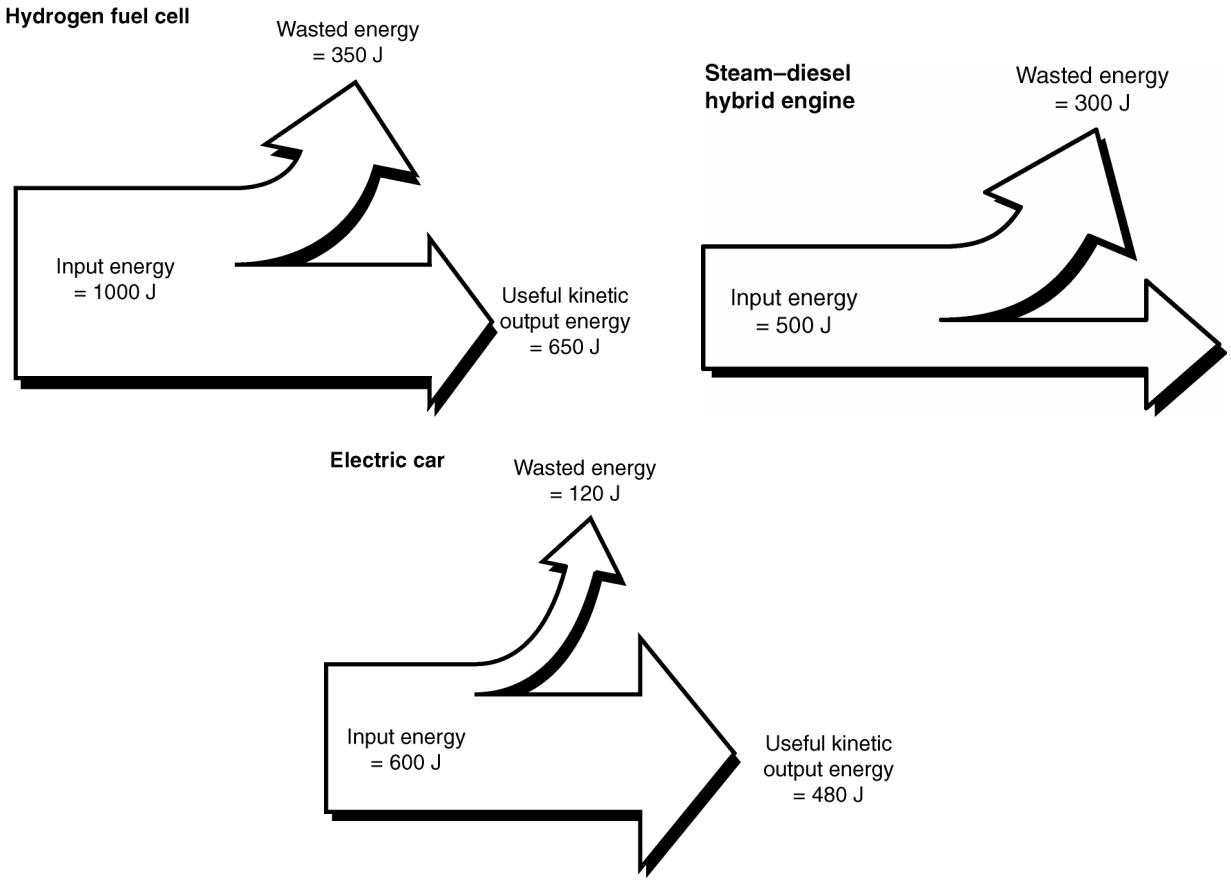
ii Machine B uses 500 J of energy to do 200 J of useful work.

4 Compare these two cranes to decide which is the more efficient.

i Crane A uses 800 J of energy to do 350 J of useful work.

ii Crane B uses 750 J of energy to do 300 J of useful work.

5 Look at these energy diagrams. Work out the efficiency of each device.


Hydrogen fuel cell
Wasted energy
= 350 J
Input energy
= 1000 J
Useful kinetic
output energy
= 650 J
Steam–diesel
hybrid engine
Wasted energy
= 300 J
Input energy
= 500 J
Electric car
Wasted energy
= 120 J
Input energy
= 600 J
Useful kinetic
output energy
= 480 J